WEEK 1 REPORT

SUB-GROUP -14

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Team Charter

SLU 06007 DVT Team 14 Team Charter

Team Members	• **Nial Jai George** (nialjaigeorge@gmail.com) • **Samkelisiwe Mhlungu** (sdlamini111@gmail.com) • **Khyati Chintha** (chintha.khyati@gmail.com) • **Siddhi Majarkhede** (majarkhedesiddhi@gmail.com) • **Laxmi Kumari** (laxmikumari4@amityonline.com) • **Soumojit Maitra** (soumojitcloud@gmail.com) • **Murali Krishna M** (muraliranikrishna@gmail.com) • **Paul Muli** (mulikituku086@gmail.com)
Team Lead	**Soumojit Maitra** (soumojitcloud@gmail.com)
Team Members Roles and Responsibilities	1. Understand the Dataset – Review the Opportunity dataset, understand its structure, identify columns and records, and become familiar with the information available. (Assigned to: Khyati Chintha) 2. Create the Data Dictionary – Document column names, their purpose, data types, and important observations to improve understanding of the dataset. (Assigned to: Laxmi Kumari) 3. Assess Data Quality – Identify issues such as missing values, duplicate records, inconsistent formatting, incorrect data types, and invalid entries. (Assigned to: Siddhi Majarkhede) 4. Clean the Dataset – Resolve the identified data quality issues by removing duplicates, standardizing values, correcting formats, and

	handling missing data. (Assigned to: Soumojit Maitra) 5. Prepare the Data Cleaning Report – Document the dataset overview, identified issues, cleaning methods applied, assumptions, and final outcomes. (Assigned to: Nial Jai George & Paul Muli) 6. Save the Cleaned Dataset – Save and organize the cleaned dataset in a proper format so it is ready for further analysis and dashboard development. (Assigned to: Murali Krishna M) 7. Review and Validate Deliverables – Check the accuracy, consistency, formatting, and alignment of the Data Dictionary, Data Cleaning Report, and Cleaned Dataset. (Assigned to: Samkelisiwe Mhlungu)
Mission, Vision Objectives & Core Values	Mission: To work together to deliver practical, high-quality recommendations that meet our sponsor's needs and can be implemented successfully. Vision Objectives: Success means working collaboratively, communicating openly, meeting deadlines, and delivering a valuable outcome for our sponsor. Core Values: • Integrity: We act honestly, ethically, and transparently in all our work. • Accountability: We take responsibility for our tasks and commitments. • Discipline: We stay organized, meet deadlines, and maintain high standards. • Respect: We value each other's ideas, communicate professionally, and embrace diversity. • Innovation: We encourage creative thinking and practical solutions to deliver value.

Internal Checks, Balances, and Reviews	Team Accountability and Performance
Accountability Mechanics	• Each team member is responsible for completing their assigned tasks on time, communicating progress, and asking for support when needed. • Sub-teams will provide brief progress updates before each full team meeting and notify the team of any challenges that may affect deadlines. • The full team will meet once a week (or more frequently if needed) to review progress, discuss challenges, assign new tasks, and ensure everyone is aligned with the project goals. • Individual performance will be measured using SMART goals, ensuring tasks are: - Specific: Clearly defined responsibilities. - Measurable: Progress tracked through completed deliverables. - Achievable: Tasks assigned according to each member's skills and capacity. - Realistic: Deadlines are practical and agreed upon by the team. - Time-based: All tasks have clear due dates.
Operations: Assignments	Understanding the dataset, creating the data dictionary, assessing data Quality, cleaning the Dataset, preparing the data, cleaning report, saving the cleaned dataset, and finally reviewing and validating deliverables.
Meetings	Team members met via Google Meet on 11 July 2026. We introduced ourselves, discussed our project or assessment to make sure everyone understood our tasks and responsibilities, and at the end we were assigned to our tasks and responsibilities.

Communication Guidelines & Status Updates	Communication Guidelines: The team lead made sure that every team member knew and understood their task; he shared updates and communicated mainly on Google Chat. Status Updates: Team members shared the project through Google Chat, and we used Google Docs for easy access to view and update the documents.
Deadlines	Project Draft is due July 12, The final project is due by July 13.

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Dataset Overview & Structure

Scope of This Section

This section documents the Opportunity dataset exactly as received from Excelerate in its original, unmodified form.

No data cleaning, preprocessing, transformation, standardization, validation, correction, or feature engineering has been performed for the purpose of this document.

The objective of this section is to provide a clear understanding of the dataset's source, purpose, structure, organization, and contents before any analytical or preprocessing activities begin.

This overview serves as a reference for understanding the original dataset and should not be interpreted as a cleaned or analysis-ready version.

Any observations regarding missing values, inconsistencies, duplicate records, placeholder values, malformed data, HTML content, JSON structures, URLs, or other data quality issues reflect the dataset exactly as it was received from Excelerate.

Data quality assessment, cleaning recommendations, preprocessing steps, exploratory data analysis (EDA), feature engineering, visualization, machine learning preparation, and business intelligence analysis are outside the scope of this section and are documented separately in the corresponding reports.

1.1 Purpose and Business Context

Each row in the dataset represents a single opportunity that can be published on the Excelerate learning platform.

The category column classifies these opportunities into different types, including Internship, Career, Competition, Course, Event, Engagement, Masterclass, Job Simulation, and a small number of additional categories.

The dataset contains information that describes various aspects of each opportunity.

Commercial details are captured through fields such as fee, currency_type, and microscholarship.

Scheduling and timeline information is stored in columns such as cohort, duration, duration_type, and last_date_to_apply.

Eligibility and application-related information is maintained through columns including eligibility, is_auto_approve, and tracking_questions.

Marketing and presentation content is represented by fields such as name, short_description, long_description, and image_link.

Recognition and engagement-related information is provided through badge, reward, and testimonial, while operational fields such as dropout_transaction, not_started_transaction, current_editor, and is_archived support internal content management and administration.

From a business perspective, the dataset serves as the foundation for managing the Excelerate opportunity catalog. It enables the organization to maintain and publish learning, career, and experiential opportunities while tracking their categories, durations, fees, eligibility criteria, and application processes. Additionally, it supports content management by recording editorial information, publication status, and update timestamps, thereby facilitating the maintenance and administration of the platform's opportunity offerings for students and early-career learners.

1.2 Data Source

The dataset was provided as a single Microsoft Excel workbook containing one worksheet (Sheet1) and was later converted to CSV format for analysis.

Based on the dataset contents—including email addresses associated with the vempower.org domain, image URLs hosted on Amazon S3 buckets such as exceleratenuserprofile and excelerate-profile-dev, and descriptive text referencing GlobalShala—the data appears to originate from the Excelerate learning platform (formerly operating under the GlobalShala name).

The presence of fields such as pk and opportunity_id suggests that this dataset represents an extract of the Opportunity entity from a larger relational database or application backend.

The dataset contains records describing learning, career, internship, competition, and event opportunities together with their associated metadata, administrative information, and application details.

No additional metadata regarding the extraction date, data refresh frequency, system version, or export process was provided with the dataset. Therefore, this document describes the dataset exactly as received from Excelerate, without making assumptions beyond the information contained within the dataset itself.

Table 1.1 — Headline dataset dimensions, measured directly from the dataset as received.

Metric	Value
Total Rows	5,733
Total Columns	33
Total Cells	189,189
File Format	Microsoft Excel Workbook (.xlsx)
Worksheets	1 ("Sheet1")
File Size on Disk	~2.19 MB

1.4 Overall Structure

Every row in the dataset represents a single Opportunity record.

The pk (partition key) column contains the constant value "Opportunity#" for all records, indicating that every row belongs to the Opportunity entity within the source system.

Each record is uniquely identified by the opportunity_id column, which follows the format "Opportunity#" followed by a unique identifier (for example, Opportunity#000000000GBD0AX3Z6VYRG7R75).

The combination of pk and opportunity_id suggests that the dataset was extracted from a NoSQL database, such as Amazon DynamoDB, where multiple entity types are commonly stored within a single table and distinguished using partition and sort keys.

Several columns contain embedded JSON objects rather than simple scalar values. These include Badge, CareerAddOn, Cohort, Eligibility, Reward, Panellist, Testimonial, DropoutTransaction, NotStartedTransaction, and tracking_questions. These JSON structures reference related entities stored elsewhere in the source system, typically through identifiers such as relation_id or other metadata. While these related entities are not provided as separate tables in this dataset, their references establish logical relationships between Opportunity records and other platform entities.

The remaining columns consist of scalar (single-valued) attributes that describe various aspects of each opportunity. These include classification information (such as category and code), commercial details (fee, currency type, and microscholarship), scheduling information (cohort, duration, duration type, and application deadline), descriptive content (name, summary, descriptions, and images), eligibility and application settings, operational metadata, and system-generated timestamps. Together, these fields provide a comprehensive representation of each opportunity while maintaining links to related entities through embedded JSON references.

Table 1.2 — Functional grouping of the 33 columns in the Opportunity dataset.

#	Functional Group	Columns	Purpose
1	Identifiers & Business Keys	pk, opportunity_id, code	Uniquely identify and reference each opportunity record.
2	Categorization & Classification	category, role, location, duration_type, currency_type	Classify an opportunity by type, associated role, delivery mode, and unit of measurement.
3	Numeric / Financial	duration, fee, microscholarship	Quantify program length, participation cost, and incentive amount.
4	Boolean Flags	is_archived, is_auto_approve	Represent yes/no business rules governing catalog

			visibility and the application workflow.
5	Date & Time	created_at, modified_at, last_date_to_apply	Record when a listing was created, last modified, and its application deadline.
6	Descriptive Text	name, short_description, long_description, summary, role_responsibility	Provide the marketing and informational content presented to prospective applicants.
7	Media & Administrative	image_link, current_editor	Store the opportunity's cover image URL and internal content management information.
8	Relational / Semi-Structured References	Badge, CareerAddOn, Cohort, Eligibility, Reward, Panellist, Testimonial, DropoutTransaction, NotStartedTransaction, tracking_questions	Store embedded JSON references to related entities such as badges, cohorts, rewards, eligibility rules, panellists, testimonials, transactions, and custom application questions.

1.6 Data Types and Feature Categories

This section describes the data type and analytical classification of each column in the Opportunity dataset.

Stored Data Type refers to how values are physically stored in the original dataset as received from Excelerate. In several cases, values representing numbers, dates, timestamps, or Boolean states are stored as text rather than their native data types.

Feature Category refers to the logical classification used for analysis and downstream data processing, grouping each column into categories such as Identifier, Categorical, Numerical, Boolean, DateTime, Text, URL, or JSON (Semi-Structured).

This classification provides a clear understanding of the role of each feature while documenting the dataset in its original, unmodified state, without applying any data cleaning, type conversion, or preprocessing.

Table 1.3 — Stored data type and feature-category classification for all 33 columns.

Column	Stored Data Type	Feature Category
pk	Text (constant value)	Identifier
opportunity_id	Text	Identifier
code	Text	Identifier
category	Text	Categorical
role	Text	Categorical
location	Text	Categorical
duration_type	Text	Categorical
currency_type	Text	Categorical
duration	Numeric, stored as text	Numerical
fee	Numeric, stored as text	Numerical
microscholarship	Numeric, stored as text	Numerical
is_archived	Text ("True" / "False")	Boolean
is_auto_approve	Text ("True" / "False")	Boolean
created_at	Numeric (Unix epoch milliseconds), stored as text	DateTime
modified_at	Numeric (Unix epoch milliseconds), stored as text	DateTime
last_date_to_apply	Numeric (Unix epoch milliseconds), stored as text	DateTime
name	Text	Text
short_description	Text	Text
long_description	Text (occasionally contains HTML markup)	Text
summary	Text	Text
role_responsibility	Text (occasionally HTML-formatted)	Text
image_link	Text (URL)	URL

current_editor	Text / JSON (mixed format)	JSON
Badge	JSON (embedded object list)	JSON
CareerAddOn	JSON (embedded object)	JSON
Cohort	JSON (embedded object list)	JSON
Eligibility	JSON (embedded object list)	JSON
Reward	JSON (embedded object list)	JSON
Panellist	JSON (embedded object list)	JSON
Testimonial	JSON (embedded object list)	JSON
DropoutTransaction	JSON (embedded object)	JSON
NotStartedTransaction	JSON (embedded object)	JSON
tracking_questions	JSON (embedded array of objects)	JSON

1.7 Feature Category Summary

Table 1.4 — Feature categories present in the dataset, with column counts (33 columns total).

Feature Category	Definition	Count
Identifier	Uniquely identifies or references a specific record.	3
Categorical	Represents a value drawn from a defined or intended set of labels.	5
Numerical	Represents a quantity; stored as text in the source file.	3
Boolean	Represents a true/false condition; stored as text in the source file.	2
DateTime	Represents a point in time, stored as Unix epoch milliseconds (text).	3
Text	Free-form descriptive or narrative content.	5

URL	A web address referencing an external resource (an image).	1
JSON (Semi-Structured)	One or more embedded objects referencing related entities or nested data; includes one mixed-format field (current_editor).	11

1.8 Key Fields

Table 1.5 — Fields that structurally identify or classify an opportunity record.

Field	Role
opportunity_id	Primary key. Uniquely identifies each opportunity record and is used to reference it elsewhere within the platform.
pk	Structural partition key. Marks every row as belonging to the "Opportunity" entity type within the source system's underlying table design (see Section 1.4).
code	Business/reference key. A short alphanumeric code (for example, "M636023") that serves as a human-readable identifier for an opportunity. The leading letter typically corresponds to the opportunity's category.
category	Primary classification key. Determines which product line (such as Internship, Career, Competition, Course, Event, Engagement, Masterclass, JobSimulation, and others) an opportunity belongs to and influences how several other columns are interpreted and used.

1.9 How the Information Is Organized

Conceptually, each row in the dataset represents a single opportunity, with its 33 columns organized into five logical layers of information that collectively describe the opportunity from both business and operational perspectives.

- Identity and Classification: The pk, opportunity_id, and code fields uniquely identify each opportunity record, while category, role, and location classify the opportunity by its type, intended role, and delivery mode.
- Commercial and Scheduling Information: Fields such as fee, currency_type, and microscholarship describe the financial aspects of the opportunity, whereas duration,

duration_type, Cohort, and last_date_to_apply provide information about the program length, scheduling, and application timeline.

- Descriptive Content: The name, short_description, long_description, summary, and role_responsibility columns contain the descriptive information presented to prospective applicants. The image_link column stores the URL of the opportunity's associated cover image.
- Administrative Metadata: Operational fields including created_at, modified_at, current_editor, and is_archived support internal content management by recording creation and modification timestamps, editorial ownership, and publication status.
- Embedded Relational References: Columns such as Badge, CareerAddOn, Cohort, Eligibility, Reward, Panellist, Testimonial, DropoutTransaction, NotStartedTransaction, and tracking_questions contain embedded JSON objects that reference related entities and application-specific information maintained elsewhere in the source system.

Since the related entities referenced within these JSON fields are not included as separate tables, the dataset is best understood as a single-entity extract from a larger multi-entity information system. It provides a comprehensive representation of Opportunity records while relying on embedded references to establish relationships with other entities managed by the platform.

This structural overview is intended to help readers understand how the dataset is organized before any data quality assessment, preprocessing, exploratory analysis, or other downstream analytical activities are performed.

Data Dictionary Overview

The Data Dictionary provides a structured description of every variable in the Opportunity dataset. It serves as a reference document to help team members understand the meaning, purpose, expected data type, and validation requirements of each column. Maintaining a comprehensive Data Dictionary improves data consistency, facilitates collaboration, and ensures accurate analysis throughout the project lifecycle.

Glossary & Business Terms

- Opportunity – A program, internship, scholarship, or event available to learners.
- Microscholarship – A financial reward awarded upon completing specific activities.
- Cohort – A group of learners participating in the same intake or batch.
- Reward – Points or incentives earned by learners.
- Archived Opportunity – An opportunity that is no longer active but retained for historical purposes.

Link of the Data Dictionary Sheet - Data dictionary week 1

Key Observations

- The dataset contains 33 variables covering opportunity details, financial information, eligibility criteria, administrative data and descriptive content.
- Several columns currently store dates and numeric values as text, requiring data type conversion during the cleaning phase.
- Some fields contain missing values that may be valid depending on business rules and should be documented rather than removed without verification.
- Unique identifier fields such as pk, opportunity_id, and code should be validated to ensure there are no duplicate records.
- Standardizing text fields (categories, roles, locations and Boolean values) will improve consistency and support accurate reporting and dashboard development.

Data Quality Assessment

1. Project Objective:

The objective of Task 3 was to assess the quality of the original Opportunity dataset by identifying data quality issues (such as missing values, JSON fields, metadata, categorical inconsistencies, numeric outliers, and placeholder entries) present in the source data. The assessment focuses on characterising the state of the uncleaned dataset so that source-level issues are transparently documented, informing both the cleaning process and subsequent analytical work.

2. Dataset Overview:

- Dataset Name: Opportunity Dataset
- Records: 5,733
- Columns: 33
- File Format: Microsoft Excel (.xlsx)
- Status: Uncleaned dataset assessed for source-level data quality

3. Assessment Checks Performed

The uncleaned dataset was examined across the following dimensions:

- Dataset dimensions
- Column names and formatting
- Data types
- Missing values
- Duplicate records
- Identifier validation
- Text formatting consistency
- HTML content detection
- JSON-like fields and metadata
- Categorical value consistency
- Numeric range and outlier detection
- Date validity
- Placeholder and test values

4. Assessment Findings

4.1 Missing Value Assessment:

High missing-value rates were observed across several columns of the source dataset: created_at, currency_type, duration_type, is_archived, location, long_description, role, role_responsibility, fee, microscholarship, last_date_to_apply.

Impact:

- Missing timestamps limit trend analysis.
- Missing currency values weaken financial comparisons.
- Missing duration data prevents accurate reporting on program timelines.

- Missing archive flags may cause confusion about record status.
- Missing location reduces geographic insights.
- Missing descriptive fields reduce qualitative reporting detail.

4.2 Duplicate Record Verification

- Duplicate rows found: 0
- Duplicate identifiers:
    - opportunity_id — unique and complete
    - code — one duplicate business code identified (see Section 4.3)

4.3 Identifier Validation

- Primary identifier (opportunity_id) is reliable — no duplicates, no missing values.

Duplicate business code — cell-level detail:

- Rows 1606 and 1607 both hold code "IZ18WQO".
- Row 1606: opportunity_id = Opportunity#00000000106K5PNHK4D7RMPZB6
- Row 1607: opportunity_id = Opportunity#00000000106K5PNHK4D7RMPZB611

The opportunity_id values differ only by a trailing "611" appended on the second record. Combined with the identical name field ("Opportunity name Assignment Review"), this pattern is consistent with an accidental copy-paste duplicate rather than a distinct business record.

4.4 Text Formatting and Column Naming

- Column names in the source dataset used mixed capitalization and spacing conventions (e.g., "Badge", "Cohort", "Testimonial" alongside snake_case names).
- Whitespace inconsistencies were observed across text columns (leading/trailing spaces, internal double spaces).
- HTML markup was present in long_description and role_responsibility.
- Categorical values displayed inconsistencies in capitalization and formatting.

4.5 JSON-like Fields, Metadata, and Row-Level Corruption

Several columns in the source dataset contained nested JSON-like objects rather than atomic values: badge, cohort, reward, eligibility, tracking_questions, panellist.

In addition, a specific row-level data corruption pattern was identified. Three rows contain URL-encoded JSON fragments spread across multiple columns, and one further row is affected in a single column. This appears to originate from a source-system export where nested JSON objects broke across column boundaries.

Affected cells:

- Row 381: E381 (currency_type), G381 (duration_type), I381 (image_link), J381 (is_archived), K381 (is_auto_approve)
- Row 435: E435, G435, J435, K435
- Row 2148: E2148, G2148, I2148, J2148, K2148
- Row 2323: I2323 (image_link only)

Examples of malformed content:

- J381: "%22maxWords%22: 500 }"
- J2148: "{ %22label%22: %22Yemen%22"

- I2323: JSON-object fragment beginning "{%22_id%22:%2260d1beaa3748301e8c6c9aa9%22..."

4.6 Image Link Column

The image_link column contains a mix of legitimate URLs and non-analytical artifacts. In addition to the corrupted rows in Section 4.5, placeholder or test values were observed such as "image", "img", "testing", random strings ("nhv", "aa", "eee"), and filler text ("Dolor laboriosam et", "Voluptatem est dolo"). These are source-system artifacts and reduce the reliability of this column for analytical use.

4.7 Categorical Inconsistencies

The uncleaned dataset displays two categorical fields with meaningful inconsistencies.

duration_type:

The field contains both singular and plural forms of the same unit (e.g., "month" / "months", "week" / "weeks", "day" / "days", "hour" / "hours", "minute" / "minutes", "year" / "years"). Two typographical errors are present: "yearssss" at G2439 and "da" at G5176.

currency_type:

"EURO" appears as a separate value from the ISO 4217 code "EUR" in 5 records (E1022, E2052, E3617, E5174, E5580).

The location field was not standardized during the Week 1 cleaning phase; it continues to contain multiple inconsistent representations of the same underlying category (e.g., "WFM," "wfm," "Work From Home," "work from home," and "WORK FROM HOME," as well as "Virtual" and "virtual," alongside the misspelling "vitrual").

4.8 Numeric Outliers

Several numeric values fall significantly outside the normal distribution of their respective columns:

- H872 (fee): 6,000,000,000,000 — implausible; median fee = 0, 99th percentile = 100.
- O1778 (microscholarship): 10,000,000 — median = 120, 99th percentile = 3,000.
- O5004 (microscholarship): 500,000,000 — extreme outlier.
- F380 (duration): 11,334 — median = 30, 99th percentile = 84.
- F838, F3140, F5508 (duration): 3,030
- F5584 (duration): 10,000

4.9 Date Validity

Timestamp columns (created_at, modified_at, last_date_to_apply) were originally stored as text and required conversion for date-based analysis. One anomalous value was identified: L1219 (last_date_to_apply): 1970-01-01 — the Unix epoch value, typically produced when a null or zero timestamp is converted to a date. 349 records show a last_date_to_apply before 2020. These may represent legitimate historical opportunities, but the pattern warrants a spot-check.

4.10 Data Type Consistency

All 33 columns in the source dataset were originally read as text (object), including timestamp fields (created_at, modified_at, last_date_to_apply) and numeric fields (duration, fee, microscholarship). These required explicit type conversion during cleaning.

5. Data Quality Observations

- Several columns contain extremely high missing values, limiting certain analyses.
- Four rows (381, 435, 2148, 2323) contain URL-encoded JSON fragments across multiple columns; these represent a specific source-level export failure.
- Nested JSON fields and metadata attributes are present in the source dataset and may require parsing or exclusion depending on analytical scope.
- The image_link column contains placeholder and test data alongside legitimate URLs.
- duration_type contains grammatical singular/plural forms and two typographical errors.
- currency_type inconsistency between "EURO" (5 records) and "EUR" is present in the source data.
- Numeric outliers in fee, microscholarship, and duration are present and require business validation.
- One Unix epoch date (L1219) present in last_date_to_apply.
- One duplicate business code exists with near-identical opportunity_ids; likely a copy-paste duplicate pending business validation.
- No duplicate rows; opportunity_id is complete and unique.

6. Source Dataset Quality Summary

Metric	Result
Records assessed	5,733
Columns assessed	33
Duplicate Rows	0
Duplicate opportunity_id	0
Duplicate Business Code	1 (near-identical IDs, likely copy-paste)
JSON-corrupted rows	4 (381, 435, 2148, 2323)
Missing Values	Present in multiple source columns
HTML Tags in Text Fields	Present in long_description and role_responsibility
Nested JSON Columns	badge, cohort, reward, eligibility, tracking_questions, panellist
Image Link Column	Contains placeholders/test values

Numeric Outliers	Present in fee, microscholarship, duration
Unix Epoch Dates	1 (L1219)
Currency Label Inconsistency	EURO vs EUR (5 records affected)
duration_type Typos	yearssss (G2439), da (G5176)
duration_type Singular/Plural	Both forms present

7. Limitations

Several columns in the source dataset contain high missing values, which will constrain aggregate reporting on those fields.

Nested JSON fields and metadata attributes were identified but not parsed during Week 1. Placeholder and test values in the image_link column are treated as source-level artifacts.

Row-level JSON corruption and numeric outliers require business or domain-level validation before any analytical use.

These limitations reflect source-level data quality issues and are not defects introduced by the cleaning process.

8. Recommendations

- Row-level JSON corruption (rows 381, 435, 2148, 2323) should be resolved by nulling the affected cells or dropping the affected rows.
- Numeric outliers in fee, microscholarship, and duration should be reviewed by a business stakeholder to distinguish data entry errors from legitimate values.
- The Unix epoch value at L1219 should be reset to null.
- The near-identical opportunity_id pair sharing the duplicate business code (rows 1606, 1607) should be reviewed to confirm whether one is an accidental duplicate.
- The image_link column should be treated as unreliable until placeholder and test values are separated from legitimate URLs.

9. Conclusion

The uncleaned Opportunity dataset was successfully assessed. Source-level issues including missing values, nested JSON fields and metadata, row-level corruption, categorical inconsistencies, numeric outliers, placeholder values, and one anomalous date have been documented with cell-level detail where applicable. The assessment provides a comprehensive record of data quality issues present in the source data, supporting the cleaning process and informing downstream analytical work.

Data Cleaning Report

Executive Summary

This report documents the data cleaning and preparation activities carried out on the Opportunity dataset during Week 1 of the Excelerate internship. The Opportunity dataset was profiled to identify data quality issues, cleaned to resolve those that fell within the Week 1 scope, and validated to confirm analytical readiness. This report describes the cleaning operations completed during the primary cleaning phase, the additional corrections applied during report preparation, the assumptions underpinning each decision, and the residual issues that remain flagged for downstream review. The final analysis-ready dataset contains 5,730 records across 20 columns and is standardized, consistent, and prepared for exploratory analysis and dashboard development in subsequent weeks.

1. Introduction

The purpose of this report is to formally document the process, methodology, assumptions, and outcomes of the Week 1 data cleaning phase. The objective of this phase was not to interpret or analyse the dataset, but to raise its quality to a standard suitable for downstream analytical work. This report should be read alongside the Data Dictionary and the Data Quality Assessment, which together comprise the Week 1 deliverables.

2. Dataset Overview

The Opportunity dataset represents a collection of opportunity records exported from the source system. Each record corresponds to a single opportunity and includes descriptive attributes, categorical classifications, timestamps, and financial fields.

Raw Dataset:

- Number of records: 5,733
- Number of columns: 33
- Column data types: all read as text

Cleaned Dataset: (Link: Opportunity_Analysis_Ready.xlsx)

- Number of records: 5,730
- Number of columns: 20
- Column data types: text (14), datetime (3), numeric (3)
- Primary identifier: opportunity_id

3. Data Quality Issues Identified in the Source Dataset

Initial profiling of the raw dataset surfaced the following categories of quality issues, which informed the cleaning methodology described in Section 4:

- Inconsistent column naming conventions across the 33 source columns.
- Whitespace anomalies across text columns.
- Embedded HTML markup in long_description and role_responsibility.
- High missing-value concentrations in several columns, some approaching 99.8%.

- One duplicate value in the secondary code field, with two records sharing the code but holding distinct primary identifiers.
- Inconsistent categorical formatting (capitalization, whitespace within values).
- Nested JSON-like objects in badge, cohort, reward, eligibility, tracking_questions, and panellist.
- Incorrect data types — all columns originally read as text, including timestamps and numeric fields.

A comprehensive assessment of these and related issues is provided in the Data Quality Assessment document.

4. Cleaning Methodology and Operations Completed

The following cleaning operations were performed sequentially, with validation after each step.

4.1 Column name standardization

All column names were normalized: converted to lowercase, leading and trailing spaces removed, internal spaces replaced with underscores, and unnecessary special characters stripped. For example, "Last Date To Apply" became "last_date_to_apply". This yielded machine-friendly, consistent identifiers throughout the dataset.

4.2 Whitespace removal

Leading and trailing whitespace was removed from every text column, and internal whitespace was normalized where it produced ambiguity.

4.3 Duplicate record and identifier validation

The dataset was checked for full-row duplicates; none were identified. The primary identifier opportunity_id was validated as fully unique with no missing values. The code field was checked for duplicates; one duplicate value was found. Because the two associated records held distinct primary identifiers, they were retained for business review rather than removed.

4.4 HTML content removal

HTML tags in long_description and role_responsibility were programmatically stripped, preserving the underlying readable text. This substantially improved the usability of these fields for text analysis and display.

4.5 Missing-value handling

Columns with extremely high missingness that offered no reliable analytical value were removed from the cleaned dataset. Columns with moderate missingness that retained business relevance were preserved, with missing values left as nulls rather than imputed.

4.6 Column pruning

Thirteen columns were removed from the final dataset: pk, Badge, CareerAddOn, Cohort, DropoutTransaction, Eligibility, NotStartedTransaction, Panellist, Reward, Testimonial, current_editor, summary, and tracking_questions. These were removed either because they contained nested structured objects requiring separate parsing outside the Week 1 scope, or because their missingness rates and low analytical relevance did not justify retention.

4.7 Text and categorical standardization

Categorical values were normalized for capitalization and internal formatting consistency, ensuring that logically identical values would group together correctly during aggregation.

4.8 Data type conversion

Timestamp columns (created_at, modified_at, last_date_to_apply) were converted from text to datetime. Numeric columns (duration, fee, microscholarship) were converted to floating-point numeric types. The remaining 14 columns were retained as text.

4.9 Row-level cleanup

Three records were removed during the cleaning process. These records were identified as invalid or non-informative under the applied cleaning rules and were excluded to preserve the analytical integrity of the dataset.

4.10 Validation loop

After each transformation, the dataset was re-checked for row and column integrity, duplicate identifiers, missing-value patterns, and dimensional consistency.

5. Additional Corrections Applied During Report Preparation

During the preparation of this report, a further review of the cleaned dataset identified a small number of residual formatting inconsistencies that fell within the Week 1 cleaning scope. The following corrections were applied to the working copy of the analysis-ready dataset, communicated to the team, and reflected in the Data Quality Assessment.

5.1 Typographical corrections in duration_type

- G2439: "yearssss" corrected to "years"
- G5176: "da" corrected to "days"

5.2 Currency label standardization

The currency_type field contained five records labelled "EURO" alongside the ISO 4217 standard "EUR". These were standardized to "EUR" for internal consistency: E1022, E2052, E3617, E5174, E5580 — "EURO" changed to "EUR"

5.3 duration_type singular and plural forms

The duration_type field contains both singular and plural forms of the same unit (e.g., "month" and "months"). Following team discussion, these forms have been retained on the basis that singular applies to a duration of one and plural applies to greater values. No further modification was made.

6. Assumptions

6.1 Missing values reflect source-level gaps

Missing entries in the retained columns are assumed to originate from the source system rather than from collection or transfer errors. Because no reliable proxy or business logic was available to justify imputation, missing values were preserved rather than filled.

6.2 The duplicate business code represents an ambiguous case pending business validation

The two records sharing the same code value hold different opportunity_id values. However, the two opportunity_ids differ only in a trailing "611" suffix on the second record, suggesting the possibility of a copy-paste duplicate. Both records were retained pending business review.

6.3 Nested JSON-like columns fall outside the Week 1 analytical scope

Columns containing nested structured objects were removed from the cleaned analytical dataset because parsing and normalizing these structures was outside the Week 1 scope. Should subsequent weeks require these attributes, the raw dataset remains available.

6.4 Highly sparse columns are unreliable for analysis

Columns with missing-value rates above approximately 85% and no clear analytical purpose were removed rather than retained.

6.5 Original business information is preserved

No cleaning operation was permitted to alter the underlying business meaning of any retained record. Cleaning operations were restricted to formatting, structural consistency, type conversion, and the removal of clearly redundant or low-value fields.

7. Validation and Final Outcomes

The cleaned dataset was subjected to the following validation checks:

- Duplicate rows: 0
- Duplicate opportunity_id values: 0
- Missing opportunity_id values: 0
- Duplicate business codes: 1 (retained, flagged for business review)
- Column names standardized: Yes
- HTML content in text fields: Removed
- Timestamp columns typed as datetime: Yes
- Numeric columns typed as numeric: Yes
- Typographical corrections applied: 2 (G2439, G5176)
- Currency label standardization applied: 5 cells (EURO to EUR)
- Dataset dimensions: 5,730 records by 20 columns

The cleaned dataset was saved as Opportunity_Analysis_Ready.xlsx and is available for the next phase of work. All records retained in the cleaned dataset preserve their original business meaning; no values were fabricated or imputed, and no business fields were altered beyond formatting, type conversion, and the standardization operations recorded above.

8. Unresolved Issues Flagged for Review

The following issues were identified in the cleaned dataset but were not modified during Week 1. Each has been documented in the Data Quality Assessment with cell-level detail and routed to the appropriate stakeholder for review or business validation.

8.1 Row-level JSON corruption

Four rows contain URL-encoded JSON fragments across multiple columns (rows 381, 435, 2148 across is_archived, is_auto_approve, duration_type, currency_type, and image_link; row 2323 in image_link only). Recommended action is either nulling the affected cells or dropping the affected rows.

8.2 Numeric outliers

Several records contain values that are significantly inconsistent with the rest of the distribution: a fee value of six trillion, microscholarship values in the hundreds of millions, and duration values in the thousands. These require business validation before any analytical use.

8.3 Unix epoch date

One record (L1219) contains 1970-01-01 in last_date_to_apply — the Unix epoch value, typically produced when a null or zero timestamp is converted to a date. Recommended action is to reset the value to null.

8.4 Potential duplicate record

The duplicate business code identified in Section 4.3 corresponds to two records (rows 1606 and 1607) whose opportunity_id values differ only by a trailing "611" suffix, suggesting a copy-paste duplicate rather than a distinct business record. Flagged for business review.

9. Limitations

9.1 Residual missingness in retained columns

Some retained columns still exhibit meaningful missingness, including is_archived (approximately 90.8%), role_responsibility (approximately 81.2%), role (approximately 66.9%), location (approximately 17.0%), and long_description (approximately 6.0%). Any analysis using these columns should account for missing values explicitly.

9.2 Duplicate business code pending validation

The retained duplicate code value requires domain validation. Depending on the outcome of that review, one of the affected records may need to be reclassified or removed in a subsequent phase.

9.3 Nested structured fields not analysed

The removal of nested JSON-like columns means that any information contained within Badge, Cohort, Reward, Eligibility, tracking_questions, and Panellist is unavailable to downstream analysis. Should any of these attributes become relevant, they will need to be revisited and parsed from the raw dataset.

9.4 Source-level data quality issues persist

High missingness in certain fields reflects underlying gaps in the source system rather than errors in the cleaning process. Improvements to source data quality lie outside the scope of Week 1.

9.5 Location standardization (Text and categorical standardization)

The location field was not standardized during the Week 1 cleaning phase; it continues to contain multiple inconsistent representations of the same underlying category (e.g., "WFM," "wfm," "Work From Home," "work from home," and "WORK FROM HOME," as well as "Virtual" and "virtual," alongside the misspelling "vitrual").

10. Conclusion

The Week 1 data cleaning phase has resulted in a standardized, validated, and analytically ready Opportunity dataset. Structural inconsistencies, formatting issues, and non-informative columns have been addressed while preserving the underlying business information. Additional formatting corrections applied during report preparation have further improved the internal consistency of the dataset. Residual issues have been documented and routed to the Data Quality Assessment and to business stakeholders for review. The cleaned dataset is suitable for Week 2 exploratory analysis, subsequent data-quality profiling, and eventual dashboard development.

Links to Supporting Documents

- **Original Dataset:** #OpportunityData
- **Cleaned Dataset:** Opportunity_Analysis_Ready.xlsx
- **Data Dictionary spreadsheet:** Data dictionary week 1